积分不等式

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1.Suppose $f(x) \in R[a,b]$ and is a concave function. then we have

$$f(\frac{a+b}{2}) \leq \frac{1}{b-a}\int_a^b f(x)\,dx \leq \frac{f(a)+f(b)}{2}$$

(**Hadamard inequality**)

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$\forall x \in [a,b],\ x = \frac{b-x}{b-a}a + \frac{x-a}{b-a}b$ so, $f(x) = f(\frac{b-x}{b-a}a + \frac{x-a}{b-a}b) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b)$ (property of the concave)

Integrate both sides of the equation, we get

$$\int_a^b f(x)\,dx \leq f(a) \times \frac{1}{b-a} \times \frac{(b-a)^2}{2} + f(b) \times \frac{1}{b-a} \times \frac{(b-a)^2}{2} = \frac{b-a}{2}[f(a)+f(b)].$$

▷

第一个不等式亦可以用定积分定义证明,请读者自证(hint:将区间2n等分再利用下凸的不等式)

If we change the condition of constraint put $f''(x) > 0$ (this situation is stronger) and we can use another common way to prove this inequality.

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put $F(x) = \int_a^x f(t)\,dt - \frac{x-a}{2}[f(a)+f(x)]$ ($x \in [a,b]$) $F(a) = 0$

$F'(x) = f(x) - \frac{f(a)+f(x)}{2} - \frac{x-a}{2}f'(x) = \frac{1}{2}[f(x)-f(a)] - \frac{x-a}{2}f'(x)$

According to the Lagrange mean theorem $\exists \quad \xi \in (a,x) \quad s.t.\, f(x)-f(a) = (x-a)f'(\xi)$

so $F'(x) \leq 0 \quad F(x) \leq 0 \quad F(b) \leq 0$ .The sencond inequality is established.

On the other hand,

$$f(x) \geq f(x_0) + f'(x_0)(x-x_0)$$

put $x_0 = \frac{a+b}{2}$,we get

$$f(x) \geq f(\frac{a+b}{2}) + f'(\frac{a+b}{2})(x - \frac{a+b}{2})$$

Integrate both sides of the equation we can get the first inequality.

▷

2. 设$f'(x) \in C[a,b], f(a) = f(b) = 0$, 证明

$$|\int_a^b f(x)\,dx| \leq \frac{b-a}{2}\int_a^b |f'(x)|\,dx$$

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$$\int_a^b f(x)\,dx = \int_a^b f(x)\,d(x-\frac{a+b}{2}) = f(x)(x-\frac{a+b}{2})\Big|_a^b - \int_a^b (x-\frac{a+b}{2})f'(x)\,dx$$
$$= -\int_a^b (x-\frac{a+b}{2})f'(x)\,dx$$

因为

$$|\int_a^b f(x)\,dx| \le \int_a^b |x-\frac{a+b}{2}||f'(x)|\,dx \quad (\text{积分绝对值不等式})$$

不等式右端等于

$$\int_a^{(a+b)/2}(\frac{a+b}{2}-x)\,|f'(x)|dx+\int_{(a+b)/2}^b(x-\frac{a+b}{2})|f'(x)|\,dx$$
$$\le \frac{b-a}{2}\int_a^{(a+b)/2}|f'(x)|\,dx+\frac{b-a}{2}\int_{(a+b)/2}^b|f'(x)|\,dx=\frac{b-a}{2}\int_a^b|f'(x)|\,dx$$

▷

3.设$f\in R[a,b]$,且$\int_a^b f^2(x)\,dx=1$,求证:

$$(\int_a^b f(x)cosx\,dx)^2+(\int_a^b f(x)sinx\,dx)^2\le b-a$$

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根据 Cauthy 不等式

$$(\int_a^b f(x)cosx\,dx)^2\le\int_a^b f^2(x)\,dx\int_a^b cos^2xdx$$
$$(\int_a^b f(x)sinx\,dx)^2\le\int_a^b f^2(x)\,dx\int_a^b sin^2xdx$$

把上下两式相加即为所求不等式。

▷

4.设$f(x)$在$[a,b]$上单调增加，证明:

$$\int_a^b xf(x)\,dx\ge\frac{a+b}{2}\int_a^b f(x)\,dx$$

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将$[a,b]$ n等分，$x_k=a+\frac{b-a}{n}k,\quad k=0,1,2...n$ 根据 Chebyshev 不等式可得

$$\frac{b-a}{n}\sum_{k=1}^n x_kf(x_k)\ge\frac{b-a}{n^2}(\sum_{k=1}^n x_k)(\sum_{k=1}^n f(x_k))=\frac{1}{b-a}(\frac{b-a}{n}\sum_{k=1}^n x_k)(\frac{b-a}{n}\sum_{k=1}^n f(x_k))$$

不等式左右两端求积分和得到:

$$\int_a^b xf(x)dx\ge\frac{1}{b-a}(\int_a^b x\,dx)(\int_a^b f(x)\,dx)=\frac{a+b}{2}\int_a^b f(x)\,dx$$

▷

注：设$f(x),g(x)$在$[a,b]$上单调增加可以得出下面的结论：

$$\int_a^b f(x)g(x)\,dx\ge\frac{1}{b-a}(\int_a^b f(x)\,dx)(\int_a^b g(x)\,dx)$$

应用:

设$f(x)$单调减少,证明:

$$\int_0^1\frac{f(x)}{1+x^2}\,dx\ge\frac{\pi}{4}\times\int_0^1 f(x)\,dx$$

5.设$f'(x) \geq 0$ 证明上题结论。

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令

$$F(x) = \int_a^x tf(t)\,dt - \frac{a+x}{2}\int_a^x f(t)\,dt \quad x \in [a,b]$$

我们有 $F(a) = 0$,且

$$F'(x) = xf(x) - \frac{1}{2}\int_a^x f(t)\,dt - \frac{a+x}{2}f(x)$$
$$= \frac{x}{2}f(x) - \frac{a}{2}f(x) - \frac{1}{2}\int_a^x f(t)\,dt \geq \frac{x-a}{2}f(x) - \frac{f(x)}{2}(x-a) = 0$$

故 $F(x) \geq 0 \quad x \in [a,b]$，因此$F(b) \geq 0$,即

$$\int_a^b xf(x)\,dx \geq \frac{a+b}{2}\int_a^b f(x)\,dx$$

▷

6.设$f'(x) \in C[a,b] \quad f(a) = 0$,证明:

$$\int_a^b f^2(x)\,dx \leq \frac{(b-a)^2}{2}\int_a^b [f'(x)]^2\,dx$$

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易知$f(x) = \int_a^x f'(t)\,dt$ 则根据 *Cauthy Inequality* 我们有

$$f^2(x) = (\int_a^x f'(t)\,dt)^2 \leq (\int_a^x 1^2\,dt)[\int_a^x [f'(t)]^2\,dt]$$
$$\leq (x-a)[\int_a^b [f'(t)]^2\,dt]$$

不等式两端同时积分即得到

$$\int_a^b f^2(x)\,dx \leq \frac{(b-a)^2}{2}\int_a^b [f'(x)]^2\,dx$$

▷

7.证明:

$$\int_0^{2\pi} \frac{sinx}{\sqrt{1+x^4}}\,dx > 0$$

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下面先证明一个引理,原题用引理即可证明
引理:

$$\text{设}f(x)\text{连续且单调减少, 那么}\int_0^{2\pi} f(x)sinx\,dx \geq 0$$
$$\text{原式} = \int_0^{\pi} f(x)sinx\,dx - \int_0^{\pi} f(\pi+x)sinx\,dx$$
$$= \int_0^{\pi} [f(x) - f(\pi+x)]sinx\,dx \geq 0 \quad (\text{请读者自行思考原因})$$

注:可以由引理推出

$$\int_{(2k-2)\pi}^{2k\pi} f(x)sinx \geq 0 \quad k=1,2,\ldots.n$$

▷

8.设$f(x)$单调减少,证明:

$$0 \leq \int_0^{2\pi} f(x)sinnx\,dx \leq 2[\frac{f(0)-f(2\pi)}{n}] \quad n \in \mathbb{N}^*:$$

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一方面

$$\int_0^{2\pi} f(x)sinnx\,dx = \frac{1}{n}\int_0^{2n\pi} f(\frac{n}{\pi})sinx\,dx$$
$$= \frac{1}{n}(\int_0^{2\pi} + \int_{2\pi}^{4\pi} + \ldots + \int_{(2n-2)\pi}^{2n\pi}) \geq 0 \quad (\text{根据上一题的引理})$$

另一方面

$$\int_0^{2\pi} f(\frac{x}{n})sinnx\,dx = \int_0^{\pi} f(\frac{x}{n})sinx\,dx + \int_{\pi}^{2\pi} f(\frac{x}{n})sinx\,dx \leq 2[f(0)-f(\frac{2\pi}{n})]$$
$$\int_{2\pi}^{4\pi} f(\frac{x}{n})sinx\,dx \leq 2[f(\frac{2\pi}{n})-f(\frac{4\pi}{n})]$$
$$\cdots\cdots$$
$$\text{累加得} \int_0^{2\pi} f(x)sinnx\,dx \leq 2[\frac{f(0)-f(2\pi)}{n}]$$

▷

学习完积分第二中值定理后下面给出另一种证法:

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原式可以化为

$$\frac{1}{n}\sum_{k=1}^{n}\int_{(2k-2)\pi}^{2k\pi} f(\frac{x}{n})sinx\,dx = \frac{1}{n}\sum_{k=1}^{n}\int_{(2k-2)\pi}^{2k\pi}[f(\frac{x}{n})-f(\frac{2k\pi}{n})]sinx\,dx(\text{这一步请读者停下来思考一下为什么})$$
$$f(\frac{x}{n})-f(\frac{2k\pi}{n})\text{式单调减少大于0的,故由积分第二中值定理}$$
$$= \frac{1}{n}\sum_{k=1}^{n}[f(\frac{2k-2}{n}\pi)-f(\frac{2k\pi}{n})]\int_{(2k-2)\pi}^{\xi} sinx\,dx, \quad \xi \in [(2k-2)\pi, 2k\pi]$$
$$\text{由于}0 \leq \int_{(2k-2)}^{\xi} sinx\,dx \leq 2, \text{故}0 \leq \int_0^{2\pi} f(x)sinnx\,dx \leq 2[\frac{f(0)-f(2\pi)}{n}]$$

▷

9. 设$f(x) \in C[0,1], 0 \leq f(x) < 1$ 证明:

$$\int_0^1 \frac{f(x)}{1-f(x)}\,dx \geq \frac{\int_0^1 f(x)\,dx}{1-\int_0^1 f(x)\,dx}$$

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求题设不等式就相当于求

$$\int_0^1 f(x)\,dx \leq [\int_0^1 (1-f(x))\,dx][\int_0^1 \frac{f(x)}{1-f(x)}\,dx]$$

我们使用定积分的定义来证明,求*Riemann*和

$$\frac{1}{n}\sum_{k=1}^{n} f(\frac{k}{n}) = \frac{1}{n}\sum_{k=1}^{n} f(x_k) = \frac{1}{n}\sum_{k=1}^{n} \frac{f(x_k)}{1-f(x_k)}(1-f(x_k)) \quad x_k = \frac{k}{n}$$

不妨设 $f(x_1) \leq f(x_2) \leq \ldots \leq f(x_n)$即将$f(x_k)$升序排序.

因为$\left\{\frac{f(x_k)}{1-f(x_k)}\right\}, \{1-f(x_k)\}$均为反序数列　　根据*Cheybyshev Inequality*有

$$\frac{1}{n}\sum_{k=1}^{n} f(x_k) = \frac{1}{n^2} \times n \times \sum_{k=1}^{n} \frac{f(x_k)}{1-f(x_k)}(1-f(x_k)) \leq \frac{1}{n^2}(\sum_{k=1}^{n} \frac{f(x_k)}{1-f(x_k)})[\sum_{k=1}^{n}(1-f(x_k)]$$

最后将不等式两端取$n \to \infty$即得到:

$$\int_0^1 f(x)\,dx \leq [\int_0^1 (1-f(x))\,dx][\int_0^1 \frac{f(x)}{1-f(x)}\,dx]$$

▷

若读者能将上述积分不等式题目掌握,那么就算初步掌握了积分不等式这块内容,积分不等式的题目千奇百怪,有兴趣的同学可以自行寻找题目,但现阶段不要"沉迷"在这块内容上,继续学习新的知识才是重中之重. 2022/7/2.